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Classification of Rockburst in Underground Projects: Comparison of Ten Supervised Learning Methods

Jian Zhou; Xibing Li; and Hani S. Mitri

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Supplemental Data

Table S1 Rockburst database for underground engineering in recent 20 years

No	Rock type	*D*/m	σ_θ/ Mpa	σ_c/ Mpa	σ_t/ Mpa	*SCF*	B_1	B_2	W_{et}	*Rockburst Classification*	References
1#	Granodiorite	200	90.00	170.00	11.30	0.53	15.04	0.88	9.00	M	Wang et al. 1998
2	Syenite	194	90.00	220.00	7.40	0.41	29.73	0.93	7.30	L	Wang et al. 1998
3	Granodiorite	400	62.60	165.00	9.40	0.38	17.53	0.89	9.00	L	Wang et al. 1998
4	Granite	300	55.40	176.00	7.30	0.32	24.11	0.92	9.30	M	Wang et al. 1998
5	Dolomitic Limestone	400	30.00	88.70	3.70	0.34	23.97	0.92	6.60	M	Wang et al. 1998
6	Granite	700	48.75	180.00	8.30	0.27	21.69	0.91	5.00	M	Wang et al. 1998
7	Quartzite	250	80.00	180.00	6.70	0.44	26.87	0.93	5.50	L	Wang et al. 1998
8	Quartz Diorite	890	89.00	236.00	8.30	0.38	28.43	0.93	5.00	M	Wang et al. 1998
9	Marble	150	98.60	120.00	6.50	0.82	18.46	0.90	3.80	M	Wang et al. 1998
10	Lead and Zinc Ore	N/A	108.40	140.00	8.00	0.77	17.50	0.89	5.00	H	Wang et al. 1998
11#	Ni Nepheline - P nepheline	N/A	57.00	180.00	8.30	0.32	21.69	0.91	5.00	M	Wang et al. 1998
12	Gneissic Granite	N/A	50.00	130.00	6.00	0.38	21.67	0.91	5.00	M	Wang et al. 1998
13#	Granitic Gneiss	N/A	62.50	175.00	7.25	0.36	24.14	0.92	5.00	M	Wang et al. 1998
14	Granite	N/A	75.00	180.00	8.30	0.42	21.69	0.91	5.00	M	Wang et al. 1998
15	Biotite angle, Flash	N/A	11.00	115.00	5.00	0.10	23.00	0.92	5.70	N	Wang et al. 1998

	Plagioclase Schist										
16	Diorite Granite	N/A	43.40	123.00	6.00	0.35	20.50	0.91	5.00	M	Wang et al. 1998
17[#]	Granite	N/A	18.80	178.00	5.70	0.11	31.23	0.94	7.40	N	Wang et al. 1998
18	Limestone	N/A	34.00	150.00	5.40	0.23	27.78	0.93	7.80	N	Wang et al. 1998
19	Granite	N/A	56.10	131.99	9.44	0.43	13.98	0.87	7.44	M	Su et al., 2010
20	Granite	N/A	54.20	134.00	9.10	0.40	0.15	0.87	7.10	M	Su et al., 2010
21	Granite	N/A	70.30	128.30	8.70	0.55	0.15	0.87	6.40	M	Su et al., 2010
22	Granite	N/A	60.70	111.50	7.86	0.54	14.19	0.87	6.16	H	Su et al., 2010
23[#]	migmatite	N/A	54.20	134.00	9.09	0.40	15.00	0.87	7.08	M	Bai et al., 2002
24	migmatite	N/A	70.30	129.00	8.73	0.55	11.40	0.87	6.43	M	Bai et al., 2002
25	Biotite granite porphyry	203	91.23	157.63	11.96	0.58	13.18	0.86	6.27	H	Zhang et al. 2010
26	Biotite granite porphyry	827	66.77	148.48	8.47	0.45	17.53	0.89	5.08	L	Zhang et al. 2010
27[#]	Biotite granite porphyry	896	51.50	132.05	6.33	0.39	20.86	0.91	4.63	M	Zhang et al. 2010
28	Biotite granite porphyry	1117	35.82	127.93	4.43	0.28	28.90	0.93	3.67	L	Zhang et al. 2010
29	Biotite limestone	1124	21.50	107.52	2.98	0.20	36.04	0.95	2.29	N	Zhang et al. 2010
30	Biotite limestone	1140	18.32	96.41	2.01	0.19	47.93	0.96	1.87	N	Zhang et al. 2010
31[#]	Biotite limestone	983	110.30	167.19	12.67	0.66	13.20	0.86	6.83	H	Zhang et al. 2010
32[#]	Biotite limestone	853	26.06	118.46	3.51	0.22	33.75	0.94	2.89	L	Zhang et al. 2010
33	Biotite granite porphyry	644	16.62	156.86	10.66	0.11	14.71	0.87	4.83	M	Zhang et al. 2012
34	Biotite granite porphyry	692	16.47	156.90	10.33	0.11	15.19	0.88	4.39	M	Zhang et al. 2012
35[#]	Biotite granite porphyry	970	16.43	157.95	11.06	0.10	14.28	0.87	4.99	H	Zhang et al. 2012
36[#]	Biotite granite porphyry	850	16.30	155.28	10.63	0.11	14.61	0.87	4.40	M	Zhang et al. 2012

37	Biotite granite porphyry	174	15.97	114.07	11.96	0.14	9.54	0.81	2.40	N	Zhang et al. 2011
38#	Biotite granite porphyry	275	19.14	106.31	11.96	0.18	8.89	0.80	2.07	N	Zhang et al. 2011
39	Biotite granite porphyry	187	12.96	117.81	11.96	0.11	9.85	0.82	2.49	N	Zhang et al. 2011
40	Biotite granite porphyry	267	31.05	147.85	11.96	0.21	12.36	0.85	3.00	M	Zhang et al. 2011
41	Biotite granite porphyry	215	29.09	138.50	11.96	0.21	11.58	0.84	2.77	N	Zhang et al. 2011
42	Biotite granite porphyry	272	32.40	140.88	11.96	0.23	11.78	0.84	2.86	L	Zhang et al. 2011
43#	Biotite granite porphyry	644	34.89	151.70	10.66	0.23	14.23	0.87	3.17	L	Zhang et al. 2011
44	Biotite granite porphyry	692	16.21	135.07	10.33	0.12	13.08	0.86	2.49	L	Zhang et al. 2011
45#	Biotite granite porphyry	970	30.56	160.83	11.06	0.19	14.54	0.87	3.63	H	Zhang et al. 2011
46#	Biotite granite porphyry	1107	19.36	113.87	4.43	0.17	25.70	0.93	2.38	L	Zhang et al. 2011
47	Biotite limestone	1205	33.15	106.94	2.98	0.31	35.89	0.95	2.15	M	Zhang et al. 2011
48	Biotite limestone	1184	9.74	88.51	2.98	0.11	29.70	0.93	1.77	N	Zhang et al. 2011
49	Biotite limestone	1373	11.75	83.96	2.98	0.14	28.17	0.93	2.15	N	Zhang et al. 2011
50	Biotite limestone	1689	39.94	117.48	2.98	0.34	39.42	0.95	2.37	L	Zhang et al. 2011
51	Biotite limestone	1606	39.82	128.46	2.98	0.31	43.11	0.95	2.40	M	Zhang et al. 2011
52	Biotite limestone	1220	46.22	140.07	2.01	0.33	69.69	0.97	3.29	L	Zhang et al. 2011
53	Biotite limestone	920	30.95	123.79	12.67	0.25	9.77	0.81	2.57	L	Zhang et al. 2011
54	Biotite limestone	785	40.99	186.30	12.67	0.22	14.70	0.87	4.10	M	Zhang et al. 2011
55	Biotite limestone	772	20.82	122.47	12.67	0.17	9.67	0.81	2.81	L	Zhang et al. 2011
56	Biotite limestone	644	36.09	164.05	12.67	0.22	12.95	0.86	3.59	M	Zhang et al. 2011
57	Clayey sandstone	N/A	7.28	52.00	3.70	0.14	14.05	0.87	1.30	N	Jia and Fan, 1991

58	Marble	N/A	9.57	99.70	4.80	0.10	20.77	0.91	3.80	N	Jia and Fan, 1991
59	Sandstone	920	34.15	54.20	12.10	0.63	4.48	0.63	3.17	L	Tang et al, 2003
60	Granite	1000	60.00	135.00	15.04	0.44	8.98	0.80	4.86	L	Yi et al. 2010
61[#]	Marble	1000	60.00	66.49	9.72	0.90	6.84	0.74	2.15	L	Yi et al. 2010
62	Migmatite	1000	60.00	106.38	11.20	0.56	9.50	0.81	6.11	L	Yi et al. 2010
63	Peridotite	1000	60.00	86.03	7.14	0.70	12.05	0.85	2.85	L	Yi et al. 2010
64	Lherzolite	1000	60.00	149.19	9.30	0.40	16.04	0.88	3.50	L	Yi et al. 2010
65	Amphibolite	1000	60.00	136.79	10.42	0.44	13.13	0.86	2.12	L	Yi et al. 2010
66	Sandstone	750	63.80	110.00	4.50	0.58	24.40	0.92	6.31	M	Yang et al. 2010
67	Dolomite	750	2.60	20.00	3.00	0.13	6.67	0.74	1.39	N	Yang et al. 2010
68	Phosphate rock	750	44.40	120.00	5.00	0.37	24.00	0.92	5.10	L	Yang et al. 2010
69	Red Shale	750	13.50	30.00	2.67	0.45	11.20	0.84	2.03	L	Yang et al. 2010
70	Sandstone	700	70.40	110.00	4.50	0.64	24.40	0.92	6.31	M	Yang et al. 2010
71[#]	Dolomite	700	3.80	20.00	3.00	0.19	6.67	0.74	1.39	N	Yang et al. 2010
72[#]	Phosphate rock	700	57.60	120.00	5.00	0.48	24.00	0.92	5.10	M	Yang et al. 2010
73	Red Shale	700	19.50	30.00	2.67	0.65	11.20	0.84	2.03	M	Yang et al. 2010
74	Sandstone	600	81.40	110.00	4.50	0.74	24.40	0.92	6.31	H	Yang et al. 2010
75	Dolomite	600	4.60	20.00	3.00	0.23	6.67	0.74	1.39	N	Yang et al. 2010
76[#]	Phosphate rock	600	73.20	120.00	5.00	0.61	24.00	0.92	5.10	M	Yang et al. 2010
77	Red Shale	600	30.00	30.00	2.67	1.00	11.20	0.84	2.03	H	Yang et al. 2010
78[#]	Limestone	510	15.20	53.80	5.56	0.28	9.68	0.81	1.92	N	Zhang and Li, 2009
79[#]	Diorite	510	88.90	142.00	13.20	0.63	10.70	0.83	3.62	H	Zhang and Li, 2009
80[#]	Iron ore	510	59.82	85.80	7.31	0.70	11.70	0.84	2.78	M	Zhang and Li, 2009
81	Skarn	510	32.30	67.40	6.70	0.48	10.10	0.82	1.10	N	Zhang and Li, 2009

82	Dolomitic limestone	225	30.10	88.70	3.70	0.34	23.97	0.92	6.60	H	Feng et al. 1994
83	Granite	375	18.80	171.50	6.30	0.11	27.22	0.93	7.00	N	Feng et al. 1994
84	Limestone	435	34.00	149.00	5.90	0.23	25.25	0.92	7.60	L	Feng et al. 1994
85	Clay sandstone	250	38.20	53.00	3.90	0.72	13.59	0.86	1.60	N	Feng et al. 1994
86[#]	Marble	100	11.30	90.00	4.80	0.13	18.75	0.90	3.60	N	Feng et al. 1994
87	Limestone	300	92.00	263.00	10.70	0.35	24.58	0.92	8.00	L	Feng et al. 1994
88	Diorite	330	62.40	235.00	9.50	0.27	24.74	0.92	9.00	H	Feng et al. 1994
89	Granite	223	43.40	136.50	7.20	0.32	18.96	0.90	5.60	H	Feng et al. 1994
90[#]	Diastatite anorthose	425	11.00	105.00	4.90	0.10	21.43	0.91	4.70	N	Feng et al. 1994
91	Mica Marble	N/A	46.40	100.00	4.90	0.46	20.40	0.91	2.00	L	Liang 2004
92	Grey-white Marble	N/A	23.00	80.00	3.00	0.29	26.80	0.93	0.85	L	Liang 2004
93	Granophyric Marble	N/A	46.20	105.00	5.30	0.44	19.70	0.90	2.30	L	Liang 2004
94	Crystal Tuff	204	35.00	133.40	9.30	0.26	14.34	0.87	2.90	L	Wang et al. 2009
95	N/A	<504	13.90	124.00	4.22	0.11	29.40	0.93	2.04	N	Qin et al. 2009
96	N/A	<504	17.40	161.00	3.98	0.14	31.40	0.95	2.19	L	Qin et al. 2009
97	N/A	<504	19.00	153.00	4.48	0.15	28.10	0.94	2.11	L	Qin et al. 2009
98	N/A	<504	19.70	142.00	4.55	0.16	27.90	0.94	2.26	L	Qin et al. 2009
99	Marble	428	18.70	81.20	10.60	0.23	7.66	0.77	1.50	N	Xu et al. 2008
100[#]	Marble	510	23.60	82.80	11.20	0.29	7.39	0.76	1.50	N	Xu et al. 2008
101	Granite Porphyry	460	28.60	123.60	11.50	0.23	10.75	0.83	2.50	N	Xu et al. 2008
102	Granite Porphyry	580	72.00	120.50	14.90	0.60	8.09	0.78	2.50	N	Xu et al. 2008
103	Diorite	460	29.80	132.20	7.80	0.23	16.95	0.89	4.60	N	Xu et al. 2008
104	Diorite	530	44.60	130.50	11.09	0.34	11.77	0.84	4.60	N	Xu et al. 2008
105	Diorite	569	66.10	135.20	10.90	0.49	12.40	0.85	4.60	L	Xu et al. 2008
106	Diorite	650	99.40	129.50	11.30	0.77	11.46	0.84	4.60	L	Xu et al. 2008
107	Dioritic Porphyrite	515	33.60	156.30	10.20	0.21	15.32	0.88	5.20	L	Xu et al. 2008

108	Dioritic Porphyrite	650	109.50	155.80	11.77	0.70	13.24	0.86	5.20	M	Xu et al. 2008
109	Magnetite	520	26.90	92.60	9.52	0.29	9.73	0.81	3.70	L	Xu et al. 2008
110	Magnetite	550	38.30	90.10	10.20	0.43	8.83	0.80	3.70	M	Xu et al. 2008
111#	Magnetite	630	83.90	95.60	8.69	0.88	11.00	0.83	3.70	L	Xu et al. 2008
112#	Granite	560	55.90	126.80	6.56	0.44	19.33	0.90	8.10	H	Xu et al. 2008
113	Granite	670	109.90	128.50	9.63	0.86	13.34	0.86	8.10	H	Xu et al. 2008
114#	Skarn	570	59.90	96.50	8.00	0.62	12.06	0.85	1.80	L	Xu et al. 2008
115	Quartz-feldspar Porphyry	600	68.00	106.80	6.10	0.64	17.51	0.89	7.20	H	Xu et al. 2008
116	Limestone	682	50.60	63.83	5.06	0.79	12.61	0.85	2.23	L	Li, S.L., 2001; Li and Xie, 2005
117	Limestone	682	50.60	85.36	4.91	0.59	17.38	0.89	3.41	L	Li, S.L., 2001; Li and Xie, 2005.
118	Lead-zinc	682	50.60	104.97	6.18	0.48	16.99	0.89	10.90	H	Li, S.L., 2001; Li and Xie, 2005.
119#	Pyrite	682	50.60	153.10	10.48	0.33	14.61	0.87	3.14	L	Li, S.L., 2001; Li and Xie, 2005.
120	Gneissic granite	490	120.80	151.60	10.10	0.80	15.01	0.88	20.00	H	Wang et al. 2009
121#	Porphyritic biotite granite	590	119.32	138.60	7.74	0.86	17.91	0.89	30.00	H	Wang et al. 2009
122	Porphyritic granite	595	95.67	127.37	10.51	0.75	12.12	0.85	30.00	H	Wang et al. 2009
123#	Monzogranite	784	114.44	174.71	14.42	0.66	12.12	0.85	10.00	H	Wang et al. 2009
124	Monzogranite	858	127.60	145.42	13.70	0.88	10.61	0.83	10.00	H	Wang et al. 2009
125	Monzogranite	951	126.41	158.03	14.32	0.80	11.04	0.83	10.00	H	Wang et al. 2009
126	Monzogranite	1170	108.53	113.37	10.43	0.96	10.87	0.83	10.00	H	Wang et al. 2009
127	Metasandstone	240	29.04	124.15	5.00	0.23	24.83	0.92	4.39	N	Zhang et al., 2002
128	Sandy slate	437	40.87	139.00	6.00	0.29	23.17	0.92	0.81	N	Zhang et al., 2002

129	Metasandstone	490	50.09	124.00	5.00	0.40	24.80	0.92	6.53	L	Zhang et al., 2002
130	Sandy slate	720	59.09	88.25	3.60	0.67	24.51	0.92	6.14	M	Zhang et al., 2002
131#	Metasandstone	620	62.13	124.00	5.00	0.50	24.80	0.92	4.62	L	Zhang et al., 2002
132	Sandy slate	470	40.90	88.25	3.60	0.46	24.51	0.92	4.61	L	Zhang et al., 2002
133	Sandy slate	220	22.93	88.25	3.60	0.26	24.51	0.92	0.81	N	Zhang et al., 2002
134	Amphibolite plagiogneiss	720	47.50	86.30	15.60	0.55	5.53	0.69	6.30	M	Liu, J.P., 2011
135	Black mica oblique gneiss	720	47.50	61.10	5.30	0.78	11.53	0.84	7.20	M	Liu, J.P., 2011
136#	Copper ore	720	47.50	99.20	7.30	0.48	13.59	0.86	8.31	M	Liu, J.P., 2011
137#	Diabase	720	47.50	91.30	14.50	0.52	6.30	0.73	21.00	M	Liu, J.P., 2011
138	Amphibolite plagiogneiss	780	67.20	86.30	15.60	0.78	5.53	0.69	6.30	M	Liu, J.P., 2011
139	Black mica oblique gneiss	780	67.20	61.10	5.30	1.10	11.53	0.84	7.20	M	Liu, J.P., 2011
140	Copper ore	780	67.20	99.20	7.30	0.68	13.59	0.86	8.31	M	Liu, J.P., 2011
141#	Diabase	780	67.20	91.30	14.50	0.74	6.30	0.73	21.00	M	Liu, J.P., 2011
142	Amphibolite plagiogneiss	840	77.00	86.30	15.60	0.89	5.53	0.69	6.30	H	Liu, J.P., 2011
143	Black mica oblique gneiss	840	77.00	61.10	5.30	1.26	11.53	0.84	7.20	H	Liu, J.P., 2011
144	Copper ore	840	77.00	99.20	7.30	0.78	13.59	0.86	8.31	H	Liu, J.P., 2011
145#	Diabase	840	77.00	91.30	14.50	0.84	6.30	0.73	21.00	H	Liu, J.P., 2011
146#	Amphibolite plagiogneiss	900	225.50	86.30	15.60	2.61	5.53	0.69	6.30	H	Liu, J.P., 2011
147	Black mica oblique gneiss	900	225.50	61.10	5.30	3.69	11.53	0.84	7.20	H	Liu, J.P., 2011
148#	Copper ore	900	225.50	99.20	7.30	2.27	13.59	0.86	8.31	H	Liu, J.P., 2011
149	Diabase	900	225.50	91.30	14.50	2.47	6.30	0.73	21.00	H	Liu, J.P., 2011
150#	Amphibolite plagiogneiss	960	274.30	86.30	15.60	3.18	5.53	0.69	6.30	H	Liu, J.P., 2011
151	Black mica oblique gneiss	960	274.30	61.10	5.30	4.49	11.53	0.84	7.20	H	Liu, J.P., 2011

152#	Copper ore	960	274.30	99.20	7.30	2.77	13.59	0.86	8.31	H	Liu, J.P., 2011
153	Diabase	960	274.30	91.30	14.50	3.00	6.30	0.73	21.00	H	Liu, J.P., 2011
154	Amphibolite plagiogneiss	1020	297.80	86.30	15.60	3.45	5.53	0.69	6.30	H	Liu, J.P., 2011
155	Black mica oblique gneiss	1020	297.80	61.10	5.30	4.87	11.53	0.84	7.20	H	Liu, J.P., 2011
156	Copper ore	1020	297.80	99.20	7.30	3.00	13.59	0.86	8.31	H	Liu, J.P., 2011
157#	Diabase	1020	297.80	91.30	14.50	3.26	6.30	0.73	21.00	H	Liu, J.P., 2011
158	Huanglong group marble	805	77.69	74.04	8.96	1.05	8.26	0.78	1.33	N	Li and Wang, 2009
159	Qixia group marble	795	77.07	78.30	6.80	0.98	11.51	0.84	3.11	M	Li and Wang, 2009
160	Skarn	635	67.18	132.20	16.40	0.51	8.06	0.78	3.97	M	Li and Wang, 2009
161	Garnet skarn	762	75.03	128.60	13.00	0.58	9.89	0.82	5.76	M	Li and Wang, 2009
162#	Quartz sandstone	851	80.54	237.20	17.66	0.34	13.43	0.86	6.38	M	Li and Wang, 2009
163	Siltstone	843	80.04	171.30	22.60	0.47	7.58	0.77	7.27	H	Li and Wang, 2009
164#	Diorite porphyry	722	72.56	304.20	20.90	0.24	14.56	0.87	10.57	H	Li and Wang, 2009
165	Limestone	900	47.56	58.50	3.50	0.81	16.71	0.89	5.00	L	Wang et al., 2004
166#	Limestone	1030	43.62	78.10	3.20	0.56	24.41	0.92	6.00	L	Kang, Y., 2006
167	Limestone	1032	47.56	80.30	3.50	0.59	22.94	0.92	5.00	M	Xia, B.W., 2006
168	Limestone	900	44.71	82.40	4.70	0.54	17.53	0.89	6.60	M	Xia, B.W., 2006
169#	Rhyolite	362	25.70	59.70	1.30	0.43	45.90	0.96	1.70	N	Zhang, J.F., 2010
170	Rhyolite	374	26.90	62.80	2.10	0.42	29.90	0.94	2.40	L	Zhang, J.F., 2010
171	Rhyolite	775	40.40	72.10	2.10	0.56	34.30	0.94	1.90	L	Zhang, J.F., 2010
172	Rhyolite	799	39.40	65.20	2.30	0.60	28.30	0.93	3.40	M	Zhang, J.F., 2010

173	Rhyolite	811	38.20	71.40	3.40	0.53	21.00	0.91	3.60	M	Zhang, J.F., 2010
174	Rhyolite	816	45.70	69.10	3.20	0.66	21.50	0.91	4.10	M	Zhang, J.F., 2010
175	Rhyolite	841	35.80	67.80	3.80	0.52	17.80	0.89	4.30	M	Zhang, J.F., 2010
176	Rhyolite	959	39.40	69.20	2.70	0.57	25.60	0.92	3.80	M	Zhang, J.F., 2010
177	Rhyolite	984	40.60	66.60	2.60	0.61	25.60	0.92	3.70	M	Zhang, J.F., 2010
178	Rhyolite	1112	39.00	70.10	2.40	0.56	29.20	0.93	4.80	M	Zhang, J.F., 2010
179	Rhyolite	981	57.20	80.60	2.50	0.71	32.20	0.94	5.50	H	Zhang, J.F., 2010
180	Rhyolite	808	55.60	114.00	2.30	0.49	49.50	0.96	4.70	M	Zhang, J.F., 2010
181	Rhyolite	799	56.90	123.00	2.70	0.46	45.50	0.96	5.20	M	Zhang, J.F., 2010
182	Rhyolite	768	62.10	132.00	2.40	0.47	55.00	0.96	5.00	M	Zhang, J.F., 2010
183#	Rhyolite	764	29.70	116.00	2.70	0.26	42.90	0.95	3.70	L	Zhang, J.F., 2010
184	Rhyolite	760	29.10	94.00	2.60	0.31	36.10	0.95	3.20	L	Zhang, J.F., 2010
185	Rhyolite	729	27.80	90.00	2.10	0.31	42.80	0.95	1.80	N	Zhang, J.F., 2010
186#	Rhyolite	724	30.30	88.00	3.10	0.34	28.30	0.93	3.00	L	Zhang, J.F., 2010
187	Rhyolite	808	55.60	114.00	2.30	0.49	49.50	0.96	4.70	M	Zhang, J.F., 2010
188	Rhyolite	1048	41.60	67.60	2.70	0.61	25.00	0.92	3.70	M	Zhang, J.F., 2010
189	Rhyolite	1074	40.10	72.10	2.30	0.55	31.30	0.94	4.60	M	Zhang, J.F., 2010
190	Rhyolite	980	58.20	83.60	2.60	0.69	32.10	0.94	5.90	H	Zhang, J.F., 2010
191	Rhyolite	839	56.80	112.00	2.20	0.50	50.90	0.96	5.20	M	Zhang, J.F., 2010
192	N/A	900	47.56	58.50	3.50	0.81	16.71	0.89	5.00	L	Ni, F.L., 2011
193	Limestone and shale	610	42.40	50.00	6.10	0.85	8.20	0.78	5.30	L	Zhang, B., 2007
194#	Limestone and shale	1010	63.60	50.00	4.00	1.27	12.50	0.85	5.30	M	Zhang, B., 2007
195#	Sandstone Slate	400	18.64	70.00	4.85	0.27	14.43	0.87	7.27	H	Zhao, X.F., 2007
196	Sandstone Slate	400	18.20	60.00	1.90	0.30	31.58	0.94	2.84	L	Zhao, X.F., 2007
197	Granite	290	53.00	147.20	7.18	0.36	20.50	0.91	5.00	M	Peng et al. 2010 &

											2009 Xu et al.
198	Bedrock	N/A	16.40	156.23	10.51	0.10	14.86	0.87	4.14	L	Sun et al., 2009
199	Bedrock	N/A	16.40	156.14	10.30	0.11	15.16	0.88	4.04	L	Sun et al., 2009
200#	Bedrock	N/A	16.40	157.34	10.55	0.10	14.91	0.87	4.26	M	Sun et al., 2009
201	Bedrock	N/A	16.40	155.63	10.42	0.11	14.94	0.87	4.20	M	Sun et al., 2009
202	Dolomitic limestone	380	12.00	30.00	5.58	0.40	5.38	0.69	5.10	N	Yu, X.Z., 2009
203	Biotite granite	660	50.28	77.30	7.65	0.65	10.11	0.82	2.47	L	Li, L., 2009
204#	Phyllic of granite	660	50.28	94.70	5.26	0.53	18.01	0.89	2.96	M	Li, L., 2009
205	phyllic of granitic cataclasite	660	50.28	59.00	5.23	0.85	11.28	0.84	0.88	N	Li, L., 2009
206	Biotite granite	630	44.80	77.30	7.65	0.58	10.11	0.82	2.47	L	Li, L., 2009
207#	Biotite granite	630	48.00	77.30	7.65	0.62	10.11	0.82	2.47	L	Li, L., 2009
208#	Biotite granite	630	53.40	77.30	7.65	0.69	10.11	0.82	2.47	L	Li, L., 2009
209#	Biotite granite	630	54.90	77.30	7.65	0.71	10.11	0.82	2.47	L	Li, L., 2009
210	Phyllic of granite	630	44.80	94.70	5.26	0.47	18.01	0.89	2.96	M	Li, L., 2009
211	Phyllic of granite	630	48.00	94.70	5.26	0.51	18.01	0.89	2.96	M	Li, L., 2009
212#	Phyllic of granite	630	53.40	94.70	5.26	0.56	18.01	0.89	2.96	M	Li, L., 2009
213#	Phyllic of granite	630	54.90	94.70	5.26	0.58	18.01	0.89	2.96	M	Li, L., 2009
214#	phyllic of granitic cataclasite	630	44.80	59.00	5.23	0.76	11.28	0.84	0.88	N	Li, L., 2009
215	phyllic of granitic cataclasite	630	48.00	59.00	5.23	0.81	11.28	0.84	0.88	N	Li, L., 2009
216#	phyllic of granitic cataclasite	630	53.40	59.00	5.23	0.91	11.28	0.84	0.88	N	Li, L., 2009
217	phyllic of granitic cataclasite	630	54.90	59.00	5.23	0.93	11.28	0.84	0.88	N	Li, L., 2009
218	Sandstone	463	12.00	85.00	3.60	0.14	23.61	0.92	1.50	N	Xiao, X.P., 2005

219	Marble sandstone	731	21.00	103.00	4.10	0.20	25.12	0.92	2.40	L	Xiao, X.P., 2005
220	Marble sandstone	1456	28.00	100.00	3.90	0.28	25.64	0.92	2.30	L	Xiao, X.P., 2005
221	Brecciated marble	1735	47.00	122.00	5.50	0.39	22.18	0.91	3.40	L	Xiao, X.P., 2005
222	Marble	2372	52.00	117.00	4.80	0.44	24.38	0.92	3.20	L	Xiao, X.P., 2005
223	Marble	1765	42.00	117.00	4.80	0.36	24.38	0.92	3.20	L	Xiao, X.P., 2005
224	Marble	1878	32.00	117.00	4.80	0.27	24.38	0.92	3.20	L	Xiao, X.P., 2005
225 #	Banded marble argillaceous limestone	630	24.00	110.00	4.40	0.22	25.00	0.92	3.00	L	Xiao, X.P., 2005
226	Argillaceous limestone	630	20.00	112.00	4.70	0.18	23.83	0.92	2.50	N	Xiao, X.P., 2005
227	Weathered and fresh welded tuff, tuff breccia and the K-feldspar granite porphyry	<768	32.80	160.00	6.60	0.21	24.30	0.92	4.60	L	Jia et al. 2013
228	Weathered and fresh welded tuff, tuff breccia and the K-feldspar granite porphyry	<768	44.80	160.00	6.80	0.28	23.60	0.92	4.90	L	Jia et al. 2013
229	Weathered and fresh welded tuff, tuff breccia and the K-feldspar granite porphyry	<768	50.90	160.00	7.50	0.32	21.30	0.91	5.30	M	Jia et al. 2013
230	Weathered and fresh welded tuff,	<768	44.80	160.00	6.70	0.28	23.80	0.92	4.80	L	Jia et al. 2013

	tuff breccia and the K-feldspar granite porphyry										
231	Weathered and fresh welded tuff, tuff breccia and the K-feldspar granite porphyry	<768	22.40	160.00	6.60	0.14	24.10	0.92	4.30	L	Wang et al. 2010
232 #	Amphibolite syenite	440	20.61	54.23	21.49	0.38	2.52	0.43	3.17	L	Zhang et al. 2007
233	Granite	275	48.00	120.00	1.50	0.40	80.00	0.98	5.80	M	Ding, et al., 2003
234	Granite	275	49.50	110.00	1.50	0.45	73.33	0.97	5.70	M	Ding, et al., 2003
235 #	Granite	275	63.00	115.00	1.50	0.55	76.67	0.97	5.70	M	Ding, et al., 2003
236	Dolomitic limestone	369	17.39	102.30	1.30	0.17	78.69	0.97	6.58	M	Jiang, L.F., 2008
237	Dolomitic limestone	369	17.02	85.09	1.30	0.20	65.45	0.97	6.14	M	Jiang, L.F., 2008
238	Dolomite	373	16.70	83.50	1.30	0.20	64.23	0.97	6.53	M	Jiang, L.F., 2008
239	Dolomite	373	17.35	86.77	1.30	0.20	66.75	0.97	3.22	M	Jiang, L.F., 2008
240	Sandy slate	374	16.87	80.33	1.30	0.21	61.79	0.97	6.92	M	Jiang, L.F., 2008
241 #	Sandy slate	374	17.08	94.90	1.30	0.18	73.00	0.97	6.91	M	Jiang, L.F., 2008
242 #	Moderately weathered granodiorite porphyry	365	61.72	92.40	8.28	0.67	11.16	0.84	5.43	L	2005 Wang et al.,2011 Li Moxiao. et al.
243 #	Fresh granodiorite porphyry	365	126.72	189.70	8.95	0.67	21.20	0.91	5.43	L	2005 Wang et al.,2011 Li Moxiao. et al.
244 #	Granite porphyry	700	57.97	125.37	7.74	0.67	21.20	0.46	2.86	L	Guo et al., 2011
245	Granite porphyry	700	57.97	96.16	3.77	0.46	16.20	0.20	2.53	L	Guo et al., 2011
246	Granite porphyry	700	57.97	70.68	4.19	0.60	25.51	0.19	2.87	L	Guo et al., 2011

N/A, Not available, “#” a) denotes testing sample by selecting stochastically.